The sirtuins are a class of enzymes that regulate various metabolic processes by removing post-translational modifications from enzymes. SIRT4 is an autosomally encoded sirtuin protein that colocalizes with the pyruvate dehydrogenase complex. It deacylates lysine (Lys) residues on certain metabolic proteins by removing methylglutaryl and hydroxymethylglutaryl units. Methylglutaryl-CoA and hydroxymethylglutaryl-CoA are intermediates in leucine (Leu) catabolism, and scientists hypothesized that SIRT4 regulates this pathway. To test their hypothesis, livers from wild-type (WT) and SIRT4 knockout (KO) mice were treated with α-keto acid derivatives of several amino acids, including the derivative produced from leucine transamination, α-ketoisocaproate (αKIC). Figures 1 and 2 show the rate of oxidation of α-keto acids and oxygen consumption in both WT and KO livers, respectively.

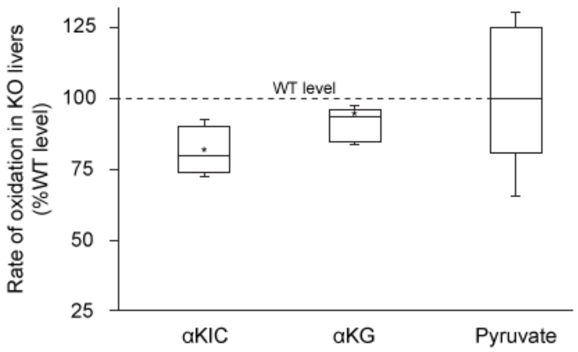


Figure 1 Box plot showing the rate of oxidation of α-ketoisocaproate (αKIC), α-ketoglutarate (αKG), and pyruvate in SIRT4 knockout livers (Note: Asterisk indicates $p < 0.05$ relative to WT level.)

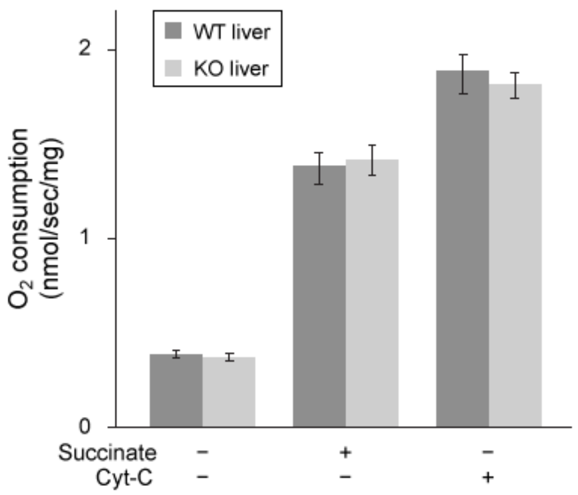


Figure 2 Oxygen consumption in WT and KO livers in the presence and absence of succinate and reduced cytochrome C (Cyt-C) (Note: Error bars represent the standard error of the mean (SEM).)

Because leucine is known to regulate insulin secretion, scientists injected WT and KO mice with leucine and measured plasma insulin levels over time. They then injected mice with insulin and monitored blood glucose levels (Figure 3).

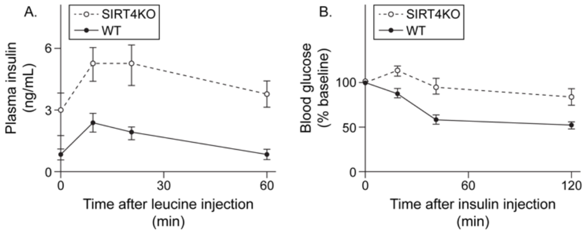


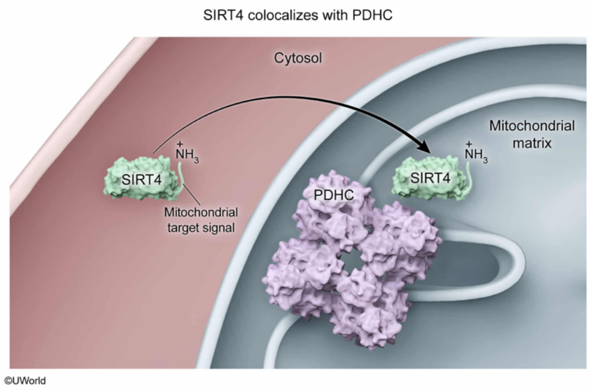
Figure 3 (A) Plasma insulin levels after injection with leucine and (B) blood glucose levels after injection with insulin in both WT and KO mice

Based on the passage, the SIRT4 protein most likely contains which of the following?

- A. A nuclear localization sequence [14%]
- B. A polyubiquitin tag [15%]
- C. An *N*-linked carbohydrate chain [12%]
- D. A mitochondrial target signal [57%]

Correct
56% answered correctly

Explanation:



In eukaryotes, **different metabolic processes** are carried out in **different compartments** of the cell. For example, glycolysis and the pentose phosphate pathway occur in the cytosol, whereas the citric acid cycle and fatty acid oxidation occur in the mitochondria. The **enzymes** required to facilitate each process must be **localized to the appropriate compartments**. Transportation of proteins synthesized in the cytosol is prompted by a **short amino acid sequence** at the **N-terminus**, which serves as a **signal** for cellular machinery to move the protein to its correct destination.

The pyruvate dehydrogenase complex (PDHC) catalyzes the decarboxylation of pyruvate to form acetyl-CoA, which then enters the citric acid cycle. This process occurs in the mitochondrial matrix, so the PDHC is found in this organelle. According to the passage, **SIRT4 colocalizes with the PDHC** and must also be found in the mitochondrial matrix. Therefore, SIRT4 most likely has a mitochondrial targeting signal to induce its transport to the mitochondria.

(Choice A) Nuclear localization sequences signal for the transport of proteins such as transcription factors to the nucleus. According to the passage, SIRT4 colocalizes with the PDHC, which is found in the mitochondria.

(Choice B) Polyubiquitin tags target defective or unnecessary proteins for destruction by the proteasome. SIRT4 is used routinely in the cell, so it would have a polyubiquitin tag only if it were defective.

(Choice C) *N*-linked carbohydrate chains are added to certain asparagine residues of proteins in the endoplasmic reticulum. Mitochondrial proteins such as SIRT4 are translated completely in the cytosol and are not modified in the endoplasmic reticulum.

Educational objective:
In eukaryotes, different metabolic pathways occur in different subcellular compartments. The enzymes required for each pathway must move to the appropriate compartment, and N-terminal peptide sequences typically direct them to their correct destination. Proteins that colocalize with one another are found in the same compartment.

Time spent:0 sec
Subject:
Biochemistry

QID:402260
System:
Metabolic Reactions

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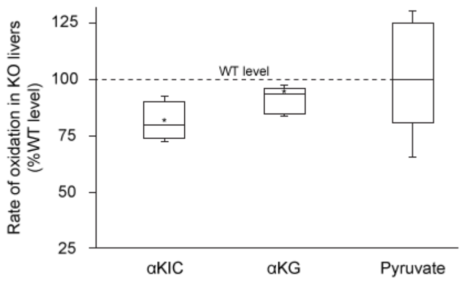


Figure 1 Box plot showing the rate of oxidation of α-ketoisocaproate (αKIC), α-ketoglutarate (αKG), and pyruvate in SIRT4 knockout livers (Note: Asterisk indicates $p < 0.05$ relative to WT level.)

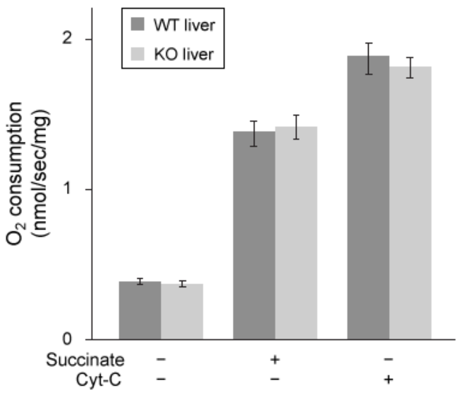


Figure 2 Oxygen consumption in WT and KO livers in the presence and absence of succinate and reduced cytochrome C (Cyt-C) (Note: Error bars represent the standard error of the mean (SEM).)

Because leucine is known to regulate insulin secretion, scientists injected WT and KO mice with leucine and measured plasma insulin levels over time. They then injected mice with insulin and monitored blood glucose levels (Figure 3).

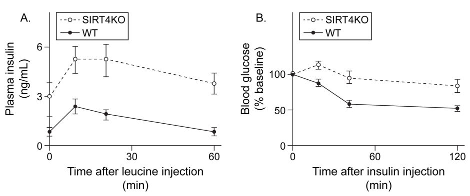


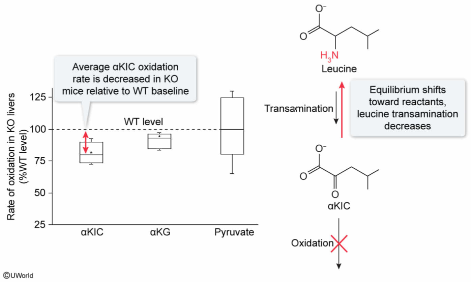
Figure 3 (A) Plasma insulin levels after injection with leucine and (B) blood glucose levels after injection with insulin in both WT and KO mice

Compared to livers obtained from WT mice, KO livers are likely to show:

- A. decreased transamination of leucine. [53%]
- B. increased deacylation of lysine. [7%]
- C. increased decarboxylation of pyruvate. [14%]
- D. decreased synthesis of acetyl-CoA. [23%]

Correct
54% answered correctly

Explanation:



The α-keto derivatives of amino acids are formed by transamination, during which the amino group is transferred to α-ketoglutarate, forming glutamate and the α-keto acid. The α-keto acid can then proceed through an **oxidative pathway**, eventually **producing acetyl-CoA** or **citric acid cycle intermediates**.

The passage states that αKIC is the α-keto derivative of leucine, meaning that it is produced by leucine transamination. The box plot in Figure 1 shows that αKIC is oxidized at a slower rate in the livers of KO mice than in the livers of WT mice. The resulting accumulation of αKIC shifts the equilibrium of the transamination reaction toward the **production of leucine (reactant)** rather than its consumption and degradation. Therefore, the transamination of leucine is likely to be **decreased** in the livers of KO mice.

(Choice B) According to the passage, SIRT4 deacylates lysine residues. Therefore, KO mice, which lack SIRT4, will most likely show *decreased* lysine deacylation.

(Choice C) The box plot in Figure 1 shows that some KO livers have higher pyruvate oxidation rates than WT livers and that some have lower rates, but the median rate is the same in both types of liver. Pyruvate oxidation occurs with decarboxylation, so pyruvate decarboxylation in KO livers is neither increased nor decreased relative to WT livers, on average. In addition, the pyruvate oxidation box plot does not display an asterisk, indicating that *p* is greater than 0.05, and that there is no statistically significant difference in pyruvate oxidation.

(Choice D) Because pyruvate is consumed at statistically identical rates in WT and KO livers, there is no change in acetyl-CoA synthesis through this pathway. In addition, multiple pathways produce acetyl-CoA, so

upregulation of one pathway may compensate for downregulation of another.

Educational objective:
Amino acids are converted into α-keto acids by transamination, in which the amino group is transferred to α-ketoglutarate. The α-keto acid can then undergo oxidation, eventually forming acetyl-CoA or citric acid cycle intermediates.

Time spent:0 sec
Subject:
Biochemistry

QID:402261
System:
Metabolic Reactions

The sirtuins are a class of enzymes that regulate various metabolic processes by removing post-translational modifications from enzymes. SIRT4 is an autosomally encoded sirtuin protein that colocalizes with the pyruvate dehydrogenase complex. It deacylates lysine (Lys) residues on certain metabolic proteins by removing methylglutaryl and hydroxymethylglutaryl units. Methylglutaryl-CoA and hydroxymethylglutaryl-CoA are intermediates in leucine (Leu) catabolism, and scientists hypothesized that SIRT4 regulates this pathway. To test their hypothesis, livers from wild-type (WT) and SIRT4 knockout (KO) mice were treated with α-keto acid derivatives of several amino acids, including the derivative produced from leucine transamination, α-ketoisocaproate (αKIC). Figures 1 and 2 show the rate of oxidation of α-keto acids and oxygen consumption in both WT and KO livers, respectively.

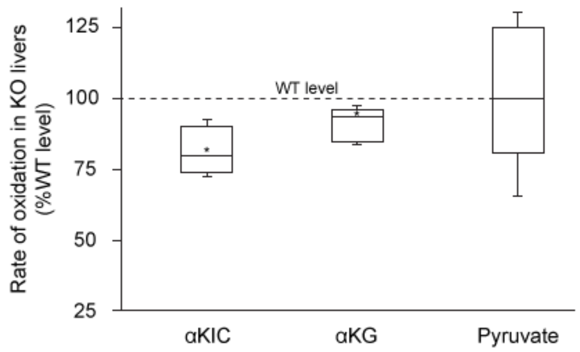


Figure 1 Box plot showing the rate of oxidation of α-ketoisocaproate (αKIC), α-ketoglutarate (αKG), and pyruvate in SIRT4 knockout livers (Note: Asterisk indicates $p < 0.05$ relative to WT level.)

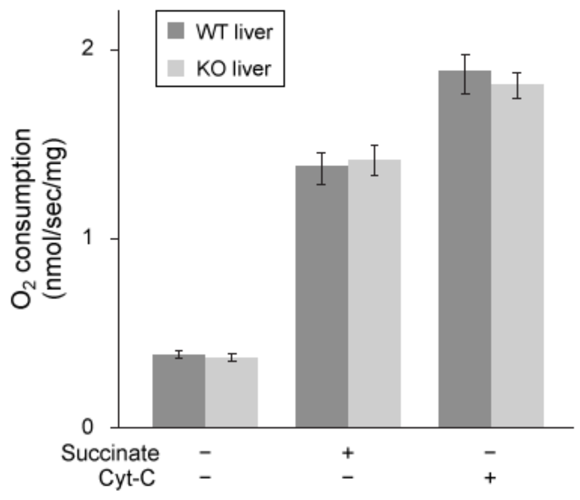


Figure 2 Oxygen consumption in WT and KO livers in the presence and absence of succinate and reduced cytochrome C (Cyt-C) (Note: Error bars represent the standard error of the mean (SEM).)

Because leucine is known to regulate insulin secretion, scientists injected WT and KO mice with leucine and measured plasma insulin levels over time. They then injected mice with insulin and monitored blood glucose levels (Figure 3).

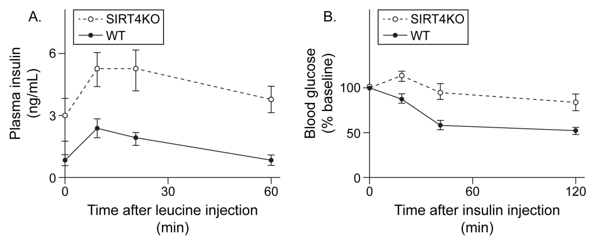


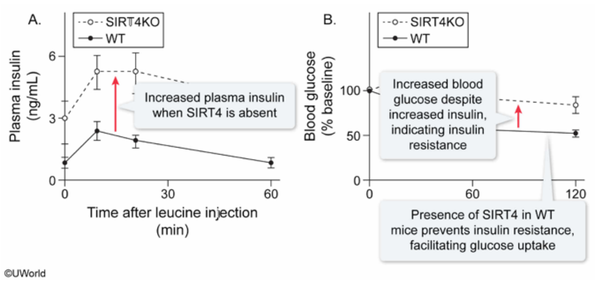
Figure 3 (A) Plasma insulin levels after injection with leucine and (B) blood glucose levels after injection with insulin in both WT and KO mice

Which conclusion about the effect of SIRT4 on blood glucose levels can be drawn from the data in Figure 3?

- A. SIRT4 causes insulin secretion, leading to decreased glucose uptake. [16%]
- B. SIRT4 facilitates glucose uptake by preventing insulin resistance. [43%]
- C. SIRT4 increases insulin sensitivity, causing increased blood glucose. [17%]
- D. SIRT4 reduces insulin secretion, inducing increased glucose uptake. [22%]

Correct
42% answered correctly

Explanation:



Metabolic signals such as high glucose or high leucine levels promote **insulin secretion by the pancreas**. Insulin then induces glucose uptake by cells, leading to a decrease in blood glucose levels. However, cells that are **overexposed to insulin** can become resistant and **take up less glucose** even when insulin is present.

Part A of Figure 3 shows that KO mice have elevated insulin levels relative to WT mice even before insulin secretion is induced by leucine injection. When leucine is injected, plasma insulin rises more in KO mice than in WT mice and takes longer to return to its baseline level. Part B of the figure shows that in WT mice, blood glucose is significantly decreased when insulin is injected. In contrast, blood glucose in KO mice remains at or above 100% of its baseline level for several minutes after insulin injection and then decreases slowly, indicating insulin resistance. Therefore, SIRT4 (present only in WT mice) **prevents the development of insulin resistance** and **facilitates glucose uptake**.

(Choice A) Figure 3 shows that WT mice produce *less* insulin than KO mice, so SIRT4 (present only in WT mice) most likely *downregulates* insulin secretion. In addition, insulin secretion would cause *increased* glucose uptake.

(Choice C) Increased insulin sensitivity would result in *increased* glucose uptake, which would lead to *decreased* blood glucose.

(Choice D) Insulin induces glucose uptake, so the decreased insulin secretion in the presence of SIRT4 (only present in WT mice) is not likely to increase glucose uptake. Instead, the absence of excess insulin likely prevents insulin resistance from developing and allows for more glucose uptake despite the lower insulin levels present in WT mice.

Educational objective:
Insulin is secreted in response to metabolic signals such as high blood glucose and induces glucose uptake by cells. Cells that are overexposed to insulin can develop resistance and take up less glucose than they normally would in the presence of insulin.

Time spent:0 sec
Subject:
Biochemistry

QID:402262
System:
Metabolic Reactions

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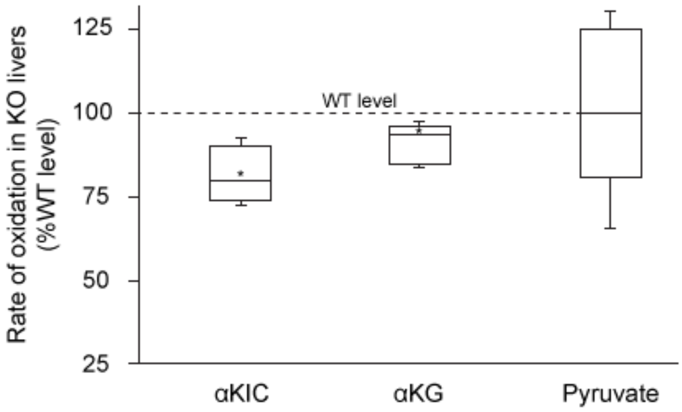
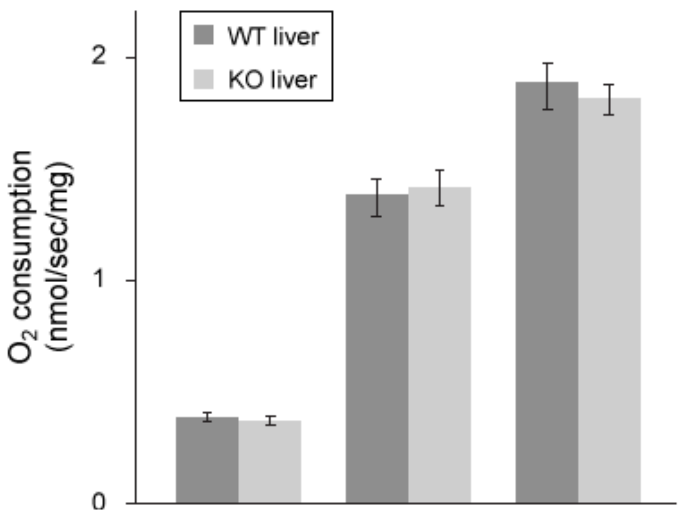


Figure 1 Box plot showing the rate of oxidation of α-ketoisocaproate (αKIC), α-ketoglutarate (αKG), and pyruvate in SIRT4 knockout livers (Note: Asterisk indicates $p < 0.05$ relative to WT level.)



Succinate – + –
Cyt-C – – +

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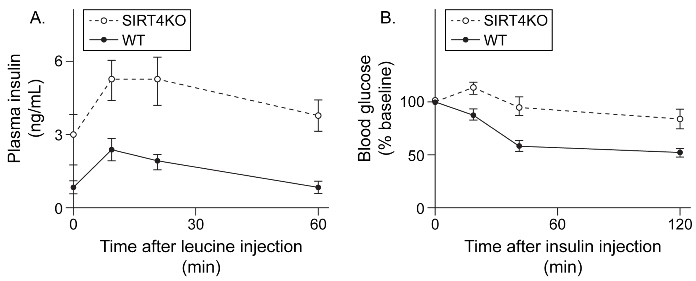


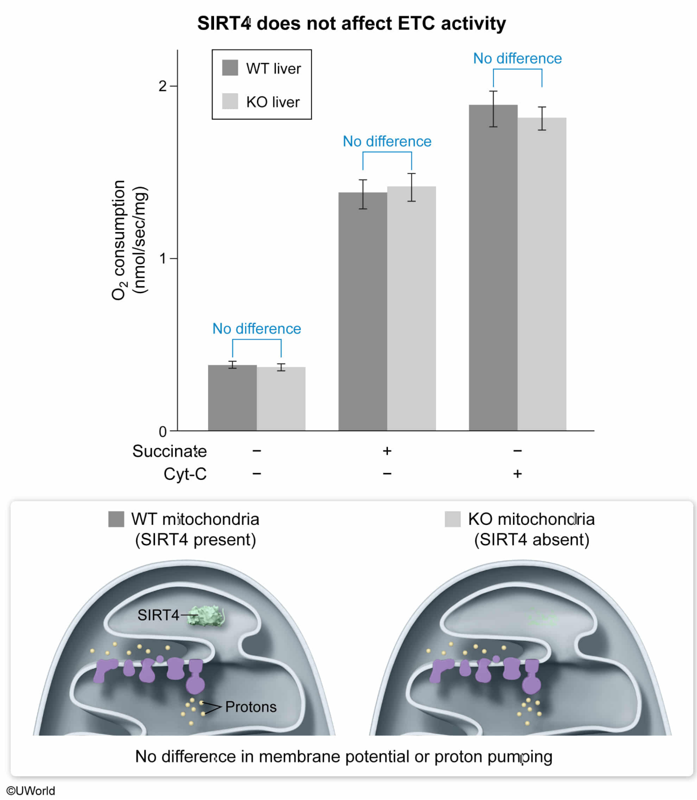
Figure 3 (A) Plasma insulin levels after injection with leucine and (B) blood glucose levels after injection with insulin in both WT and KO mice

In the absence of SIRT4, the electrochemical potential across the inner mitochondrial membrane:

- A. is increased relative to mitochondria in the presence of SIRT4. [8%]
- B. is decreased relative to mitochondria in the presence of SIRT4. [19%]
- ✔ C. is not changed relative to mitochondria in the presence of SIRT4. [66%]
- D. is increased relative to mitochondria with SIRT4 only when cytochrome C is added. [4%]

Correct
66% answered correctly

Explanation:



The electron transport chain (ETC) **pumps protons** out of the mitochondrial matrix, creating an **electrochemical potential** across the inner mitochondrial membrane as protons accumulate in the intermembrane space. These protons flow **back into the matrix through ATP synthase**, and the process reaches a **steady state** as the ETC pumps protons out at the same rate that ATP synthase allows them back in. The consumption of oxygen (O_2), which is the final electron acceptor in the ETC, can be used to monitor the rate of proton pumping. **Disruptions** in the amount of O_2 consumed by the ETC indicate a change in the steady state, which would change the **membrane potential**.

In Figure 2, researchers measured how the presence (WT) or absence (KO) of SIRT4 altered the consumption of O_2 in a variety of settings. The error bars in the image represent standard errors of the mean (SEM) for each dataset, and overlapping SEM error bars indicate that no statistical difference exists between two sets of data. In each case, the error bars overlap between WT and KO livers, indicating O_2 was consumed at essentially the same rate in both WT and KO livers. This indicates that the ETC functions normally in KO mice. Therefore, KO mouse liver cells pump protons out of the matrix and let them back in at the same rate as

WT cells, and no change in the electrochemical gradient across the inner mitochondrial membrane occurs.

(Choices A and B) Figure 2 shows that O_2 consumption is not affected by SIRT4, indicating that the ETC is equally efficient in both WT and KO livers. Therefore, protons are being pumped out of the mitochondrial matrix and traveling back in through ATP synthase at the same rate in both cases, and the electrochemical gradient is neither increased nor decreased.

(Choice D) Although Figure 2 shows that Cyt-C increases the rate of O_2 consumption, its effect is the same in both WT and KO livers.

Educational objective:
The electron transport chain pumps protons out of the mitochondrial matrix, and ATP synthase lets them back in. Healthy mitochondria reach a steady state in which proton concentration is higher outside than inside the mitochondrial matrix, producing an electrochemical potential across the membrane. Disrupting the steady state alters the electrochemical potential.

Time spent:0 sec	**QID**:402263
Subject:	**System:**
Biochemistry	Metabolic Reactions

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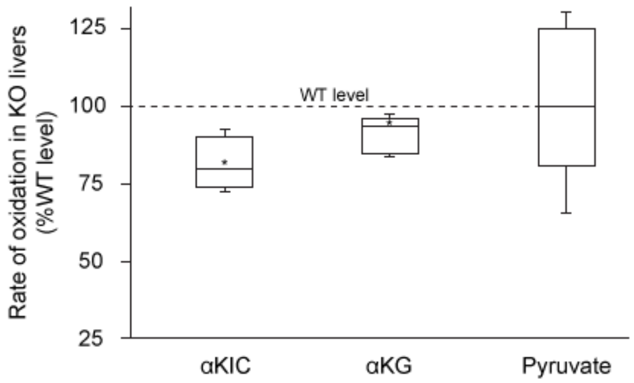
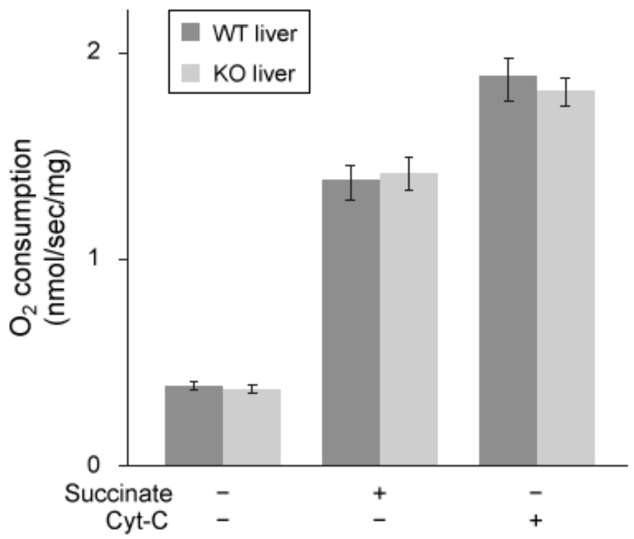


Figure 1 Box plot showing the rate of oxidation of α-ketoisocaproate (αKIC), α-ketoglutarate (αKG), and pyruvate in SIRT4 knockout livers (Note: Asterisk indicates $p < 0.05$ relative to WT level.)



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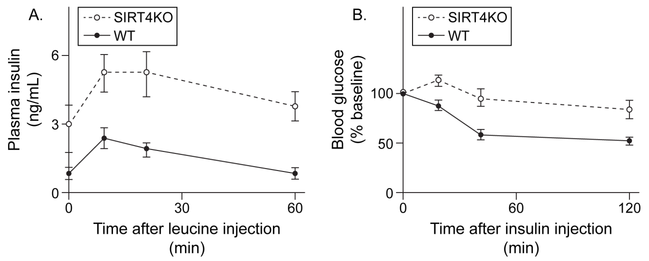


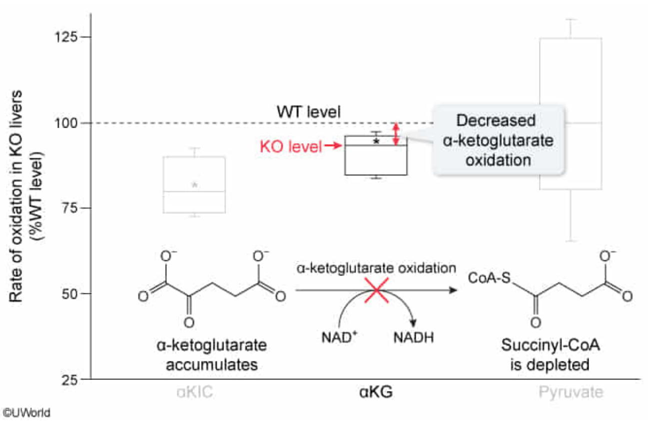
Figure 3 (A) Plasma insulin levels after injection with leucine and (B) blood glucose levels after injection with insulin in both WT and KO mice

Under aerobic conditions, which of the following molecules would be expected to be present in significantly decreased levels in the livers of KO mice?

- A. Succinyl-CoA [44%]
- B. ATP [22%]
- C. Isocitrate [20%]
- D. Cytochrome C [12%]

Correct
43% answered correctly

Explanation:



The citric acid cycle is an **oxidative process** in which a six-carbon molecule (citrate) is converted into a four-carbon molecule (oxaloacetate). The number of carbons is reduced by two successive **oxidative decarboxylation** events in which carbon dioxide (CO_2) is released. The second event, referenced in the

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passage as the oxidation of α-ketoglutarate, requires α-ketoglutarate dehydrogenase (αKDH). This enzyme removes a CO_2 from α-ketoglutarate (five carbons) and forms a bond between a carbon atom and coenzyme A to produce succinyl-CoA, a four-carbon molecule.

Figure 1 shows that α-ketoglutarate is oxidized at a significantly lower rate in KO mice than in WT mice. Because all livers were treated with the same amount of α-ketoglutarate, this indicates that αKDH works more slowly in KO mice than in WT mice under identical experimental conditions. Therefore, the product of α-ketoglutarate oxidation, **succinyl-CoA**, will most likely be **depleted** in KO livers.

(Choice B) Figure 2 shows that oxygen consumption is not affected by SIRT4, indicating that the electron transport chain and oxidative phosphorylation are fully functional. Because this is the primary means of ATP generation under aerobic conditions, it is unlikely that ATP levels are significantly reduced in KO mice.

(Choice C) Isocitrate is a precursor to α-ketoglutarate in the citric acid cycle. Because α-ketoglutarate is consumed more slowly in KO livers, isocitrate is also likely to be consumed more slowly, and it will *accumulate*.

(Choice D) Figure 2 shows that KO mouse livers consume oxygen at the same rate as WT livers, indicating that the Cyt-C necessary to reduce oxygen is present at the same level in both genotypes.

Educational objective:
The citric acid cycle facilitates several oxidative reactions, including two oxidative decarboxylation events that use isocitrate and α-ketoglutarate as substrates, respectively. Inhibition of a step in the cycle leads to accumulation of reactants and depletion of products.

Time spent:0 sec
Subject: Biochemistry

QID:402264
System: Metabolic Reactions

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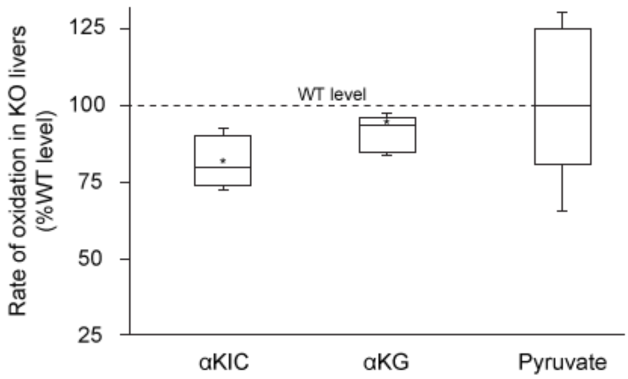

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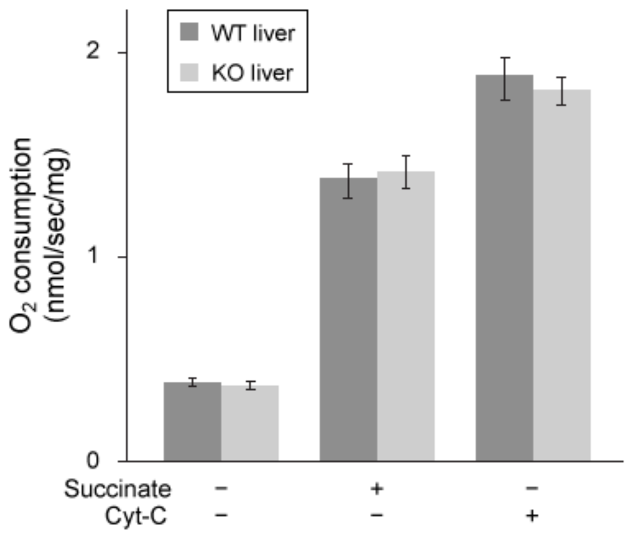

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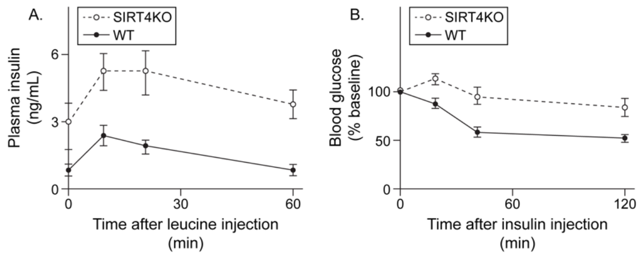


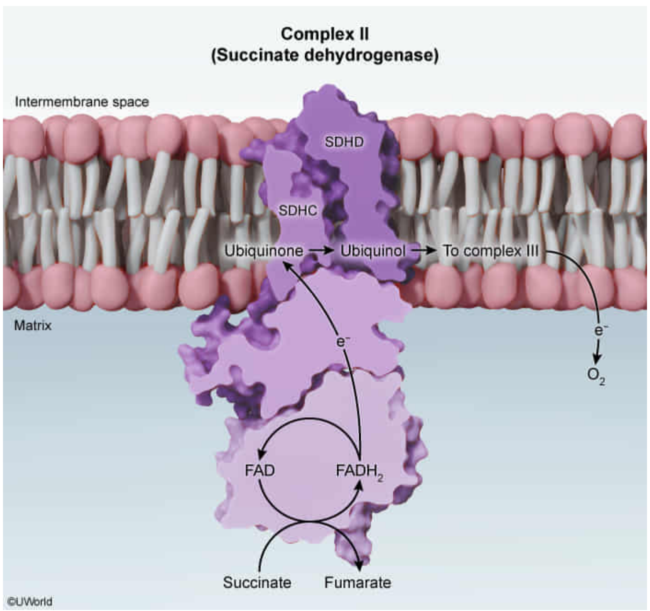
Figure 3 (A) Plasma insulin levels after injection with leucine and (B) blood glucose levels after injection with insulin in both WT and KO mice

In Figure 2, the most likely reason that researchers added succinate was to stimulate:

- A. NADH synthesis by Complex I. [4%]
- B. GTP synthesis by succinyl-CoA synthetase. [21%]
- C. $FADH_2$ formation by Complex II. [66%]
- D. ubiquinol synthesis by fumarase. [6%]

Correct
67% answered correctly

Explanation:



The electron transport chain (ETC) depends on the production of **NADH and**

$FADH_2$ by the citric acid cycle and other metabolic pathways. **Complex II** is a flavoprotein (an oxidoreductase that contains an FAD or FMN prosthetic group) that is *also known as* **succinate dehydrogenase**. It is **part of both the ETC and the citric acid cycle**. In Complex II, electrons are transferred from **succinate** to an FAD molecule that is covalently attached to the protein complex, **producing fumarate and $FADH_2$**. The $FADH_2$ then transfers its electrons to ubiquinone, forming ubiquinol and regenerating FAD.

Oxygen is the final electron acceptor in the ETC and is consumed only if sufficient NADH and $FADH_2$ are present. Increasing the rate of $FADH_2$ synthesis increases the rate of electron transfer and therefore increases the rate of oxygen consumption. In Figure 2, researchers measured oxygen consumption to assess ETC output under various conditions. They most likely added succinate to **stimulate formation of $FADH_2$** (and the subsequent electron transfer) **by Complex II**.

(Choice A) The addition of succinate can indirectly stimulate NADH synthesis at the downstream step catalyzed by malate dehydrogenase. However, Complex I only *consumes* NADH; it does not produce it.

(Choice B) Succinyl-CoA synthetase produces GTP only when it consumes succinyl-CoA to produce succinate. Adding succinate would shift the equilibrium to favor the reverse reaction and would not stimulate GTP synthesis.

(Choice D) Ubiquinol is formed in Complex I and Complex II of the ETC. It is not produced by fumarase.

Educational objective:
The electron transport chain (ETC) depends on the production of NADH and $FADH_2$ to function. Complex II of the ETC is a flavoprotein (contains FAD) that is also known as succinate dehydrogenase. It is part of both the citric acid cycle and the ETC, and it oxidizes succinate to reduce FAD to $FADH_2$. $FADH_2$ is then oxidized back to FAD in the same enzyme, producing ubiquinol.

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